# Public Key Crypto notes - 01

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## 1 Symmetric key crypto recap

- Symmetric key algorithms, schemes (AES, DES, block ciphers, MACs, etc.)
- Given two parties A, B
- They have a *shared secret key* $k$
- There is an encryption function $Enc_k(m) = c$
- And there's also a decryption function $Dec_k(c) = m$
- Properties:
  - Symmetric: the roles, keys of A and B are symmetric
  - Fast and efficient encryption & decryption

## 2 Public key (asymmetric) cryptography (PKC)

- Encryptions (RSA, ElGamal, elliptic curve based cryptography)
- Digital signatures
- Syntax of PKC:
  - There are two kinds of keys
  - $e$ - Encryption key, which is public
  - $d$ - Decryption key, which is private, kept in secret
- $c = Enc_e(m)$
- $m = Dec_d(c)$

### 2.1 Properties of PKC

- Asymmetric
- Slower, computationally more expensive calculations compared to symmetric key cryptography

- Since it is so expensive to individually encrypt and decrypt each message in an asymmetrical way, modern ciphersuites often do the following:
  - Exchange the shared secret key between two parties using each party's keypairs (e.g. using the Diffie-Hellman key exchange),
  - Once the shared key is exchanged in an encrypted way over any channel, continue the communication using a symmetric cipher (like AES in GCM mode) and the shared key (which is the same for both parties).
- Ralph Merkle: "The inventor of PKC" and commutative encryption.

**Definition 2.1** (Commutative encryption)**.**

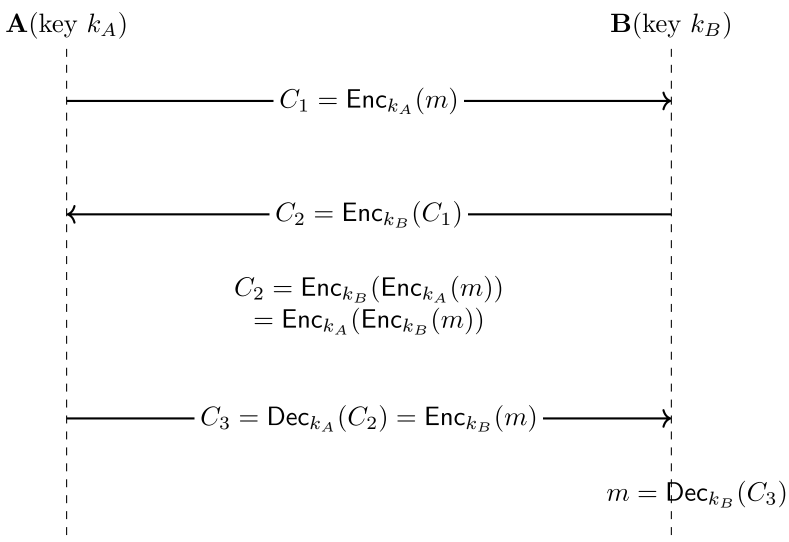
An encryption scheme $\mathsf{Enc}_k(\cdot)$ is *commutative* if for all messages $m$ and all keys $k_1, k_2$,

$$\mathsf{Enc}_{k_1}(\mathsf{Enc}_{k_2}(m)) = \mathsf{Enc}_{k_2}(\mathsf{Enc}_{k_1}(m)).$$

Equivalently, the order of encryption under different keys does not matter:

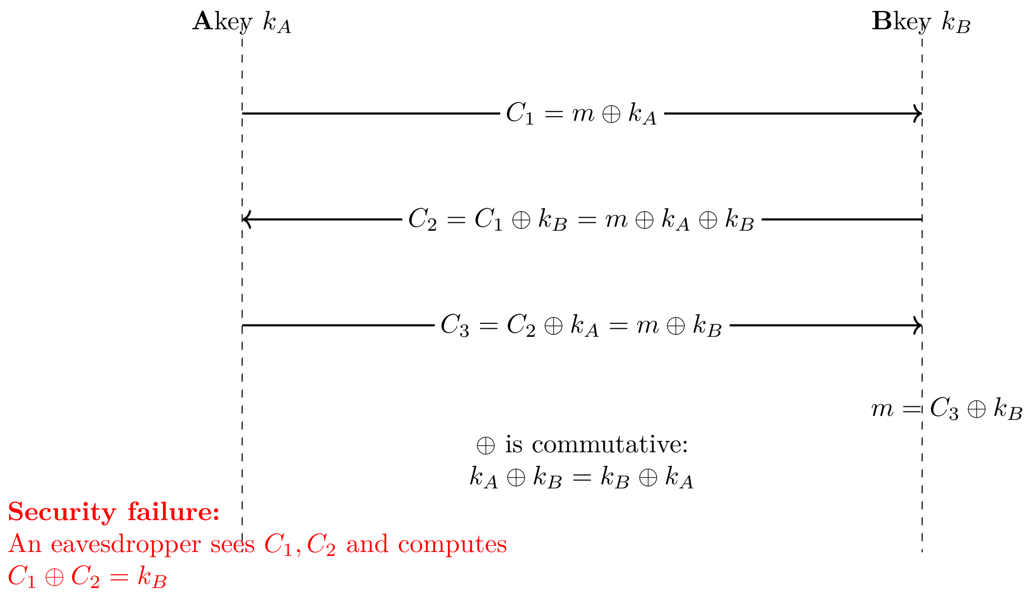
$$\mathsf{Enc}_{k_1} \circ \mathsf{Enc}_{k_2} = \mathsf{Enc}_{k_2} \circ \mathsf{Enc}_{k_1}.$$

## 2.2 R. Merkle - exchanging a shared key using a commutative encryption scheme




- This way, the shared key never gets sent on the insecure channel as a plaintext.
- It always travels with at least one layer of encryption.
- Let's see if the one-time pad scheme could be used this way.

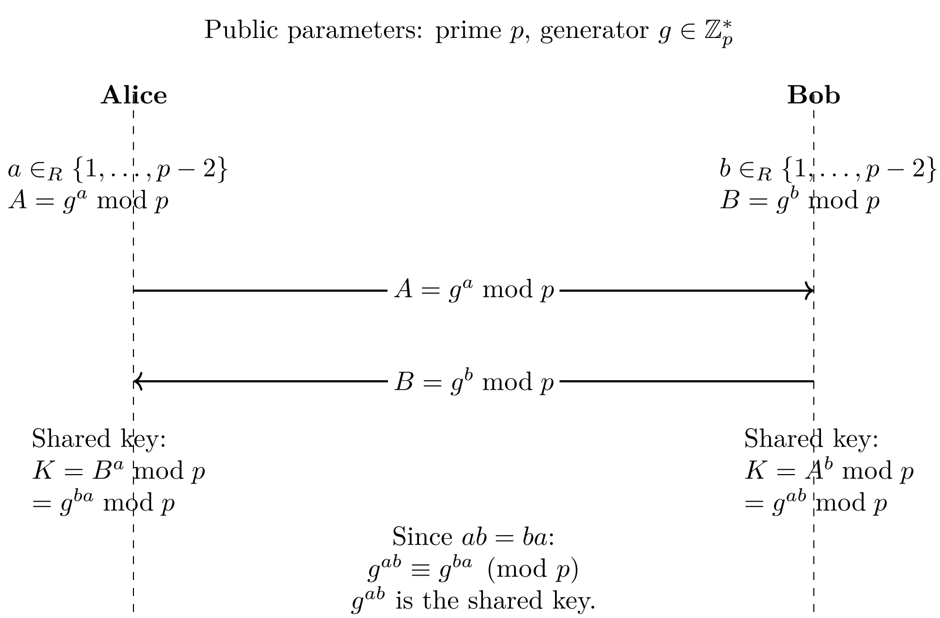
**One-Time Pad as Commutative Encryption (Broken Example)**




- The OTP scheme could not be used this way, since B's key would leak in the presence of an eavesdropper.

## 2.3 Diffie-Hellman (DH) key exchange

- Whitfeld Diffie and Martin Hellman developed a secure commutative encryption scheme in 1973 and is mainly used as a key exchange protocol.
- Take a large prime $p$ and a number $g \pmod p$, where $g \in \mathbb{Z}_p$




- If $a$ can be computed from $g^a$. then this key exchange protocol would be breakable.

- This problem is the *Discrete logarithm problem*, which is computationally hard.
- Questions:
- Is the Discrete logarithm problem really hard?
- What is the generator number $g$? How can we choose it?
- How large would $p$ need to be?

# 3 Number theoretic backgrounds

**Definition 3.1** (Order of the generator)**.** Let $p$ be a prime number and $g \in \mathbb{Z}_p, g \neq 0$. Then, the order of $g$ ($ord_p g$) is the smallest $k \leq 1$, such that:

$$g^k \equiv 1 \pmod p$$

**Remark 3.1.** This is well-defined by Fermat's little theorem: $g^{p-1} \equiv 1 \pmod p$.

**Remark 3.2.** $ord_p g$: The order of $g$ is the number of possible powers of $g$ modulo $p$.

$$g^0 = 1; g^1 = g; g^2; g^3; \dots g^{ord_p(g)-1} \tag{1}$$

$$g^{ord_p(g)} = 1; g^{ord_p(g)+1} = 1 \cdot g = g \tag{2}$$

**Claim 3.1.** $ord_p(g) | p-1$ ($ord_p(g)$ divides $p-1$)

*Proof.* Let $\sigma = ord_p(g)$

$$p - 1 = \sigma \cdot g + r \qquad 0 \leq r \leq \sigma \tag{3}$$

$$1 = g^{p-1} = g^{\sigma \cdot g + r} = \qquad \text{(by Fermat's little theorem)} \tag{4}$$

$$= (g^\sigma)^g \cdot g^r \qquad \text{(where } g^\sigma \text{ comes from the definition of the order of } g\text{)} \tag{5}$$

i.e.: $1 = g^r$ by the minimality of the $ord$, $r = 0$ □

**Definition 3.2** (generator)**.** Let $p$ be a prime and $g \in \mathbb{Z}_p, g \neq 0$. Then $g$ is a generator if:

$$ord_p(g) = p - 1$$

**Remark 3.3.** One can prove that there are always generators.

**Theorem 3.1.** Let $p$ be a prime and write $p - 1 = q_1^{e_1} \cdot q_2^{e_2} \cdot \dots \cdot q_t^{e_t}$. Then $g$ is a generator if and only if:

$$g^{\frac{p-1}{q_i}} \equiv 1 \pmod p$$

for all $i$.

*Proof.* Prove from left-to-right: We know that $g$ is a generator. Then:

$$ord_p(g) = p - 1$$

So, by the minimality of the $ord$, $g^{\frac{p-1}{q_i}} \not\equiv 1 \pmod{p}$.

Prove from right-to-left: Suppose that $g$ is *not* a generator, i.e.: $g^k \equiv 1 \pmod{p}$, $1 \leq k < p-1$. As $k|p-1$, write $p-1 = s \cdot k$. Let $q_i|s$, then:

$$g^{\frac{p-1}{q_i}} = g^{\frac{s \cdot k}{q_i}} = g^{\frac{s}{q_i} \cdot k} = (g^k)^{\frac{s}{q}} = 1$$

□

**Remark 3.4.** Typical choice for $p$ is $p = 2q + 1$, where $q$ is a prime. Here, $p$ is called a "safe" prime, while $q$ is called a "Sophie Germain prime".[1]

[1] https://en.wikipedia.org/wiki/Safe_and_Sophie_Germain_primes